Human Agreement Is the Ceiling: Evaluating Seven Frontier LLMs as Rubric Graders Across Two Engineering Courses

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Abstract

Large language models are increasingly proposed as graders for open-response engineering coursework, but the published evidence is contradictory: reports range from automated graders outperforming certified human re-graders to models falling far below human inter-rater agreement on the same task. Most of this evidence is produced by comparing a model against a *single* reference grader — a design that silently assumes human graders agree with one another. We show that this assumption largely determines the conclusion.

We evaluate seven frontier language models (GPT-4o, GPT-5, Gemini 2.5 Pro, Gemini 2.5 Flash, and Claude Haiku 4.5, Sonnet 4.6, and Opus 4.6), each run three times, on two engineering assignments graded independently by three human graders each: a 133-point Operating Systems assignment (40 students, 240 question instances) and a 10-point Biomaterials assignment (5 students, 25 question instances). This yields 945 model gradings against 755 human question-level judgements.

We propose and apply a **human-ceiling protocol**: quantify inter-human reliability first, form a consensus reference from it, and then ask whether inserting the model as an additional rater preserves the panel's reliability. Three findings follow. First, the protocol is not cosmetic — on the Operating Systems assignment, a conventional single-reference analysis reverses the *sign* of measured grading bias for five of seven models and reports significant bias ($p < 0.001$) for two models that a corrected analysis finds indistinguishable from the human consensus. Second, no model reaches human variability in either course: the best model's disagreement with the human panel is 2.80x the humans' disagreement with each other on Operating Systems and 1.76x on Biomaterials, and inserting any model as a fourth rater lowers ICC(2,1) in both courses (-0.091 to -0.209, and -0.077 to -0.161, respectively). Third, the error is not primarily random: twelve of fourteen model-course pairs show statistically significant *harshness*, and the offset accounts for most of the gap — which makes it correctable.

Because our two courses differ sharply in human ceiling (ICC 0.956 vs 0.504) while sharing one protocol and one model set, we reproduce the literature's contradictory verdicts within a single study, and argue that much of the field's disagreement concerns reference quality rather than model capability.

We conclude that autonomous LLM grading is not defensible at current capability, but that per-rubric bias calibration plus human review of flagged items is. We argue the field should report inter-human reliability as a precondition for any claim about LLM grading accuracy.

Keywords. automated grading, large language models, inter-rater reliability, assessment, engineering education, LLM evaluation

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1. Introduction

Grading open-response engineering work is expensive, slow, and — as every instructor who has moderated a teaching team knows — inconsistent. The arrival of capable large language models (LLMs) has therefore prompted a wave of proposals to automate it.

The published results do not agree. Kortemeyer reports $R^2 = 0.84$ against human graders on introductory physics derivations [4]; Bernik et al. report Pearson 0.91 on programming assignments [6]; Gobrecht et al. report an automated grader whose median absolute error is 44% *smaller* than that of certified human re-graders [3]; Tang et al. report human-AI agreement comparable to human inter-rater reliability on physics exams [9]. Against these, Mathew et al. report all tested models below QWK 0.30 where human raters reach 0.72 [10], Caraeni et al. judge GPT-4o's accuracy on handwritten mathematics ”too low for real-world settings” [11], and Lundgren finds low inter-rater reliability between GPT-4 and instructors even when average grades match [12]. These are not differences of degree; they are opposite conclusions from competent studies.

Most of this evidence shares a design choice that we believe explains much of the disagreement. A model's output is compared against one reference — a single instructor, a single teaching assistant, or an answer key — and agreement with that reference is reported as accuracy. The design is convenient, and for objective item types it is sound. For rubric-scored open response it embeds an assumption that is rarely tested: that the reference grader *is* the correct grade, and by extension that a second human grader would have produced substantially the same number.

Flodén, comparing ChatGPT against the teachers who set three Master's-level exams, states the problem precisely without being able to resolve it: having found that the model's grades differed from the teachers' in most cases, he observes that ”it is not unlikely that two different human graders could result in similar discrepancies” [5]. Resolving that requires several independent human graders on the same items — which most studies, including that one, do not have.

That assumption is measurable, and when we measure it we find it does not hold uniformly. Across our two courses, the same three-grader protocol produced inter-rater ICC(2,1) of 0.956 in one course and 0.504 in the other. In the first, human graders are nearly interchangeable and a demanding bar for an LLM is appropriate. In the second, the humans themselves disagree substantially, and a model judged against any one of them would be scored largely on which human it happened to be compared against. A single number — ”correlation with the instructor” — cannot distinguish these two situations, yet they call for opposite deployment decisions.

This paper makes four contributions.

A method. We describe a *human-ceiling protocol* for evaluating LLM graders (Section 3). It requires measuring inter-human reliability before any model is scored, constructing a consensus reference rather than privileging one grader, and then applying an insertion test: does adding the model to the human panel as an additional rater preserve the panel's reliability? The protocol converts the vague question ”is the model accurate?” into the answerable question ”is the model within the variability the department already tolerates among its own graders?”

Evidence. We apply the protocol to seven frontier models across two dissimilar engineering assignments, with three independent human graders per assignment and three repeated runs per model (Sections 4–6). The repeated runs let us separate a model's systematic error from its run-to-run instability, a property invisible to single-run studies but decisive for deployment.

Guidance. We show that the dominant error component is a systematic harshness offset rather than noise, and we derive from this a set of deployment recommendations that are neither ”replace the TAs” nor ”do not use LLMs” (Section 8).

A reconciliation. Because our two courses have very different human ceilings while sharing one protocol and one set of models, we can reproduce the field's contradictory verdicts within a single study (Section 8.5). We argue that much of the disagreement in the literature is a disagreement about reference quality rather than about model capability, and we state the prediction that would test this.

We also report, as a cautionary result, what the conventional analysis would have concluded on the same data (Section 5.7). The difference is not marginal: five of seven models change the sign of their reported bias, and the ranking of which models are ”significantly biased” substantially reorders. We present this not to criticise prior work but to argue that the human ceiling is a precondition for interpreting any of these numbers.

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2. Related Work

2.1 Automated grading before LLMs

Automated assessment has a long history in engineering and computing education. Burrows, Gurevych, and Stein's survey of automatic short answer grading (ASAG) identifies 35 systems across five temporal eras, tracing the field from concept-mapping and information-extraction approaches through corpus-based and machine-learning methods [1]. Their component analysis established what has remained the field's most durable finding: performance is strongly item-dependent, and the era of *evaluation* — rather than of new architectures — is what consolidates the field. Our results in Section 5.6, where per-question MAE varies by a factor of five across items within a single assignment, are consistent with this.

The pre-LLM ceiling was set by fine-tuned transformers. Dada et al. released a structured dataset of open-ended university responses and found Longformer best among traditional and transformer approaches, at Spearman 0.77 against human graders [2]. Most notably for our purposes, Gobrecht et al. trained an ASAG model on university exam data and evaluated it against *certified human domain experts re-grading historic exams*, reporting that the model's median absolute error was 44% smaller than the human re-graders' — concluding that automated grading is more consistent than humans and can therefore increase fairness [3]. This is the strongest pro-automation claim in the literature we review, and it is explicitly ceiling-referenced. We return to it in Section 8.5.

2.2 LLMs as graders in higher education

Instruction-tuned LLMs removed the need for per-item training data and prompted a rapid wave of evaluations in higher education. **The reported results are contradictory, and that contradiction is this paper's point of departure.**

Favourable findings. Kortemeyer graded handwritten introductory-physics derivations with MathPix and GPT-4, achieving $R^2 = 0.84$ against human graders, but concluded that the workflow suits formative feedback and that for final evaluations it ”would best be used to assist human graders” [4]. Flodén conducted the study closest in design to ours: three Master's-level exams, 463 responses, each graded at least three times by ChatGPT for 1,389 total gradings, compared against the teachers who set them [5]. Seventy per cent of gradings fell within 10% of the teachers' marks and 31% within 5%. Bernik et al. compared GPT-4-turbo, Claude 3, and Gemini 1.5 Pro on 315 Python assignments, with GPT-4 achieving Pearson 0.91 and mean absolute deviation 0.68 points [6]. Poličar et al. concluded that, with well-designed prompts, LLM grading of a bioinformatics course was comparable to human teaching assistants in both scoring and feedback [7]. Henkel et al. reported GPT-4 at Cohen's kappa 0.70 against human raters at 0.75 on K-12 short-answer marking [8]. Tang, Ambrose, and Cheng scored constructed-response physics exams using four instructors across two rounds and found human-AI agreement on total scores *comparable to human inter-rater reliability*, though the model struggled specifically with mid-range responses involving partial or ambiguous reasoning [9].

Unfavourable findings. Mathew et al. evaluated six models on the ASAP and DREsS essay corpora and found all below QWK 0.30 against a human-human ceiling of QWK 0.72, reporting that LLMs compress scoring ranges — inflating underdeveloped essays while penalising minor language errors in otherwise strong work [10]. Caraeni, Scarlatos, and Lan evaluated GPT-4o on handwritten university mathematics exams and found that although rubrics improved alignment, overall accuracy remained ”too low for real-world settings” [11]. Lundgren found that GPT-4 produced comparable *average* grades to human instructors while its inter-rater reliability with those instructors was low, and that it graded risk-aversely, attending to surface features rather than disciplinary standards [12].

Broader reviews of AI-assisted grading document both the efficiency case and the persistent need for human oversight [13], and instructor acceptance remains an independent constraint on adoption [14].

Documented failure modes include systematic leniency or severity [10], [12], compression toward the centre of the scale [5], [10], sensitivity to prompt formulation — where semantically equivalent prompts shift behaviour substantially [15] — and instability across repeated invocations of an identical prompt. Stureborg et al. characterise LLM evaluators as both inconsistent and biased, reporting skewed rating distributions, anchoring effects, and low agreement with themselves on identical samples [16]. This last property motivates our repeated-run design (Section 5.6); it is invisible to single-run studies.

2.3 LLM-as-judge and its evaluation designs

A parallel NLP literature evaluates LLMs as judges of model output rather than of student work. Zheng et al. introduced MT-Bench and Chatbot Arena and documented position, verbosity, and self-enhancement biases in LLM judges [17]. Notably, they benchmark GPT-4's agreement with humans (approximately 85%) against *human-human* agreement (approximately 81%) — precisely the ceiling-referenced comparison we argue educational studies should adopt as standard. The insertion test of Section 3 is adapted from this tradition of treating the judge as one rater among several rather than as an oracle.

2.4 Inter-rater reliability in educational assessment

Measurement theory has long held that a single rater is an unreliable instrument. Shrout and Fleiss formalised the intraclass correlation coefficient and its six forms, distinguishing the reliability of a single rater from that of a k-rater average [18]; Cohen's weighted kappa extends chance-corrected agreement to ordinal scales with partial credit [19]; Hayes and Krippendorff argue for alpha as a standard reliability measure, notably because it tolerates missing judgements [20] — which real grading data has, as Section 5.1 demonstrates. Generalizability theory, developed by Cronbach et al. [21] and applied to performance assessment by Brennan [22], decomposes measurement error into facets including raters, items, and occasions, and is the natural framework for the question we ask.

2.5 The gap

Ceiling-referenced comparison is not new, and we do not claim to have introduced it. Gobrecht et al. [3], Henkel et al. [8], Tang et al. [9], Mathew et al. [10], and Zheng et al. [17] all report human agreement alongside model agreement. Our contribution is narrower.

Consider what the literature above collectively asserts. Gobrecht et al. report a model beating human re-graders by 44% [3]; Tang et al. report parity with human inter-rater reliability [9]; Mathew et al. report QWK below 0.30 against a human 0.72 [10]. These are not differences of degree. **They are opposite answers to the same question, produced by competent studies using defensible statistics.**

Flodén states the resolving hypothesis explicitly but does not test it. Observing that ChatGPT's grades differed from the teachers' in most cases, he notes that ”it is not unlikely that two different human graders could result in similar discrepancies” [5]. That is exactly the right question — and answering it requires multiple independent human graders on the same items, which that study, like most, did not have. The hypothesis has been available for years without being measured.

Our contribution is therefore threefold.

1. **The insertion test.** Reporting a human ceiling *beside* a model score is weaker than asking whether the model, added to the existing panel, preserves that panel's reliability. We are not aware of this test being applied to LLM grading of engineering coursework, and it is the question a department actually faces when it considers deployment.
2. **A demonstration that the design choice changes the answer, on one dataset.** Section 5.7 shows that the conventional single-reference, run-pooled analysis reverses the sign of measured bias for five of seven models on our data, and reverses significance verdicts in both directions. This converts ”you should measure the ceiling” from methodological advice into an empirical result.
3. **Two courses, one protocol, ceilings differing by a factor of nearly two.** This lets us address the contradiction above directly (Section 8.5) rather than adding a further data point to it.

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3. The Human-Ceiling Protocol

The protocol has six steps. Steps 1–2 concern the humans and must be completed before any model is scored; steps 3–6 concern the models.

Step 1 — Quantify the human ceiling

With three or more independent graders scoring the same items, compute at the *question-instance* level:

- **ICC(2,1)** — reliability of a single randomly chosen rater; the quantity that matters when one grader will grade one submission.
- **ICC(2,k)** — reliability of the k-rater average; the quality of the consensus itself.

- **Krippendorff's alpha** (interval) — tolerates missing judgements, which real grading data has.
- **Kendall's W** — concordance of the rank ordering the graders impose on students.
- **Mean pairwise human-human MAE**, in points per question — the interpretable form. This is the number instructors actually reason about.

These jointly define the ceiling. The last is the most useful for the deployment question, because it answers ”how many points apart are two of our own graders, on average?”

Step 2 — Form a consensus reference

Rather than privileging one grader, take the per-question **mean** across graders as the reference, with the median computed as a robustness check. Where the two diverge, the item is contested and should be flagged rather than averaged away. Missing judgements are skipped in the mean, not imputed — but the resulting asymmetry must be tracked, because it propagates (Section 5.1).

Step 3 — Score each model against the consensus

Compute MAE, RMSE, Pearson r, Spearman rho, agreement-R^2, bias (signed mean error), and error SD. Compute these **per run and then average across runs**, rather than pooling runs, so that repeated measurements of the same item are not treated as independent observations.

Step 4 — The insertion test

Insert the model into the panel as an additional rater and recompute ICC(2,1) and Krippendorff's alpha over the enlarged panel. Compare to the human-only baseline. If reliability is preserved, the model is contributing rater-quality judgements; if it drops, the model is injecting disagreement. This is the protocol's central test, because it asks the deployment question directly: *would we accept this as one more grader on the team?*

Step 5 — The within-human-variability test

Compute each model's mean absolute disagreement with each individual human, and compare it to the mean human-human disagreement. A model whose disagreement with the humans is no larger than the humans' disagreement with each other is, by the department's own operative standard, an acceptable grader. This yields the interpretable **ceiling ratio**:

> ceiling ratio = (model's mean MAE vs. individual humans) / (mean human-human MAE)

A ratio of 1.0 means the model is exactly as far from the humans as they are from each other. Anything above 1.0 is a model that disagrees with the staff more than the staff disagree among themselves.

Step 6 — Run-to-run self-consistency

Human graders grade once; a model can be invoked repeatedly. For every (student, question, model) triple, compute the standard deviation of scores across runs and average. This measures grading *noise* — a distinct failure mode from bias, invisible to run-pooled analysis, and directly relevant to fairness: a noisy grader assigns different grades to identical work.

Why the order matters. Steps 1 and 2 must precede steps 3–6. Once a single grader has been designated ”the reference”, every subsequent number inherits that grader's idiosyncrasy, and there is no way to recover the ceiling after the fact.

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4. Study Design

4.1 Grading pipeline

All models were driven through a single grading service that presents each model with an identical task structure: for each question, the question text, an optional reference answer, the rubric, the maximum points, and the student's answer; and requests a per-question score with strengths, areas for improvement, and a breakdown. Questions are flattened — multi-part questions are expanded into their sub-parts — and each submission is graded in a single bulk call rather than one call per question, so the model sees the whole submission in context.

Objectively-scorable item types (multiple choice, true/false) are graded deterministically by the service rather than by the model, and are excluded from all analyses reported here. Every number in this paper therefore concerns open-response, rubric-scored items only.

Three provider APIs were used: the OpenAI Chat Completions API, Amazon Bedrock's Converse API for the Anthropic models, and the Google GenAI API for the Gemini models.

4.2 Models

Model	Provider / API	Decoding setting
GPT-4o	OpenAI	temperature 0.1, max 16384 tokens
GPT-5	OpenAI	reasoning effort ”high” (temperature not applicable)
Claude Haiku 4.5	Anthropic via Bedrock	temperature 0.1, max 2000 tokens
Claude Sonnet 4.6	Anthropic via Bedrock	temperature 0.1, max 2000 tokens
Claude Opus 4.6	Anthropic via Bedrock	temperature 0.1, max 2000 tokens
Gemini 2.5 Flash	Google GenAI	temperature 0.1, max 2000 tokens
Gemini 2.5 Pro	Google GenAI	temperature 0.1, max 2000 tokens

Each model graded every submission **three times**. These decoding settings are not fully uniform across providers, and we treat that as a limitation rather than a design feature; see Section 9.

4.3 Course 1 — Operating Systems (primary study)

An undergraduate Operating Systems assignment: 6 open-response questions, 133 points total, with per-question maxima ranging from 15 to 40 points. Forty student submissions were graded independently by three teaching assistants (TA1, TA2, TA3), and by each of the seven models three times. Student answers were supplied to the models as structured text.

This yields 240 question instances, 680 human question-level judgements (720 less the 40 missing TA2 Q6 scores, Section 5.1), and 5,040 model question-level scores (240 x 7 models x 3 runs).

4.4 Course 2 — Biomaterials (contrast study)

A Biomaterials assignment: 5 open-response questions, 2 points each, 10 points total. Five student submissions were graded independently by three human graders, and by each of the seven models three times. Submissions were supplied to the models as **PDFs ingested directly**, rather than as extracted structured text.

This yields 25 question instances, 75 human judgements, and 525 model scores.

On the role of this study. With five students, this arm is underpowered and we do not treat it as independent confirmation of the Operating Systems results; its confidence intervals are wide and overlapping (Section 6). It is included because its human ceiling is *low* — ICC(2,1) = 0.504 against 0.956 for Operating Systems — which makes it the more informative case for the protocol itself. It demonstrates that the bar an LLM must clear is a property of the course and its graders, not a universal constant.

4.5 Analysis

Both courses were analysed with the identical protocol of Section 3. Confidence intervals are bootstrap (percentile) intervals. Paired comparisons between a model and the human consensus use the run-averaged score per student as the unit, giving n = 40 and n = 5 respectively, and are reported with both paired t-tests and Wilcoxon signed-rank tests alongside Cohen's d. Rank correlations across the two courses are Spearman's rho over the seven models.

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5. Study 1 — Operating Systems

5.1 A data-quality finding that changes the results

Before reporting agreement, one property of the human data must be stated, because it propagates into every naive comparison.

TA2 did not grade Question 6. All forty of TA2's Q6 scores are missing. TA2's recorded total is therefore a sum over Q1–Q5 (out of 93) while TA1's and TA3's are sums over Q1–Q6 (out of 133). Any analysis that averages the three recorded totals to form a reference produces a reference that is approximately 8.8 points too low — because one third of the average is missing an entire question worth 40 points.

The consequence is that every model appears roughly 8.8 points *more generous* than it is. As Section 5.7 shows, this is enough to reverse the sign of the reported bias for five of the seven models. We handle it by computing the consensus per question instance (so the Q6 consensus is the TA1/TA3 mean), by computing ICC and Kendall's W on the fully-crossed Q1–Q5 subset, and by using Krippendorff's alpha — which tolerates missingness — on all 240 instances.

We report this in detail because it is not an exotic failure. Partial grading, split grading duties, and missing rubric rows are ordinary features of real course data, and a single-reference pipeline absorbs them silently.

5.2 The human ceiling

Table 1 — Inter-human agreement, Operating Systems.

Measure	Value
ICC(2,1), single rater (200 fully-crossed units)	**0.956**
ICC(2,k), 3-TA average	0.985
Krippendorff's alpha, interval (240 units)	0.956
Kendall's W	0.969
Mean pairwise TA-TA MAE	**1.113 pts/question**
— TA1–TA2 / TA1–TA3 / TA2–TA3	1.10 / 1.53 / 0.70
Mean pairwise quadratic-weighted kappa (grade bins, Q1–Q5)	0.896
Mean pairwise TA-TA MAE, total score (Q1–Q5, /93)	3.46 pts
TA mean totals (Q1–Q5, /93)	TA1 56.5, TA2 55.6, TA3 53.9

These TAs are close to interchangeable. ICC(2,1) = 0.956 means a single randomly chosen TA already reproduces the panel almost exactly, and the three TAs' mean totals span 2.6 points out of 93. **The bar for a model in this course is therefore high, and it is high for a principled reason, not because of a convention that ICC should exceed 0.9.**

5.3 Models against the consensus

The mean and median consensus agree closely (r = 0.998); we use the mean.

Table 2 — Question-level metrics vs. the TA consensus (240 instances, per-run averaged). Reference: TA-TA MAE = 1.113 pts/question.

Model	MAE	RMSE	Pearson r	Spearman rho	R^2	Bias	Error SD
Claude Opus	**3.148**	**4.691**	**0.879**	**0.818**	**0.753**	−0.719	**4.645**
Claude Sonnet	3.316	4.853	0.872	0.811	0.736	−1.077	4.742
Gemini 2.5 Pro	3.594	5.739	0.831	0.747	0.628	**−0.343**	5.739
Gemini 2.5 Flash	3.725	5.813	0.829	0.750	0.621	−0.411	5.807
Claude Haiku	3.945	6.192	0.804	0.762	0.570	−2.253	5.779
GPT-5	3.956	6.145	0.810	0.721	0.577	−0.904	6.090
GPT-4o	4.891	6.868	0.766	0.652	0.472	−1.595	6.691

Read on its own, this table looks like a success. Correlations of 0.77–0.88 are in the range routinely reported as evidence that LLMs can grade. **The MAE column tells a different story.** The best model is 3.148 points per question away from the consensus, against a human-human disagreement of 1.113 — nearly three times as far. Correlation is high because the models rank students correctly; the absolute grades are not close.

Every bias is negative. All seven models grade this assignment harshly.

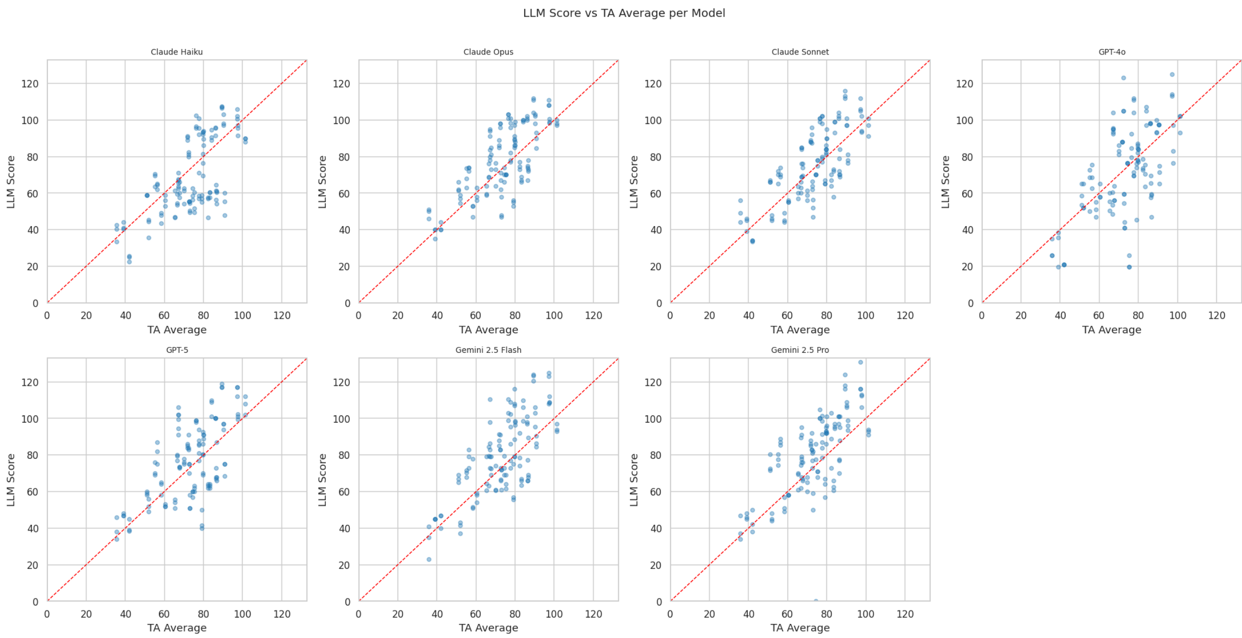

Figure 1. per-model scatter of model score against TA consensus.

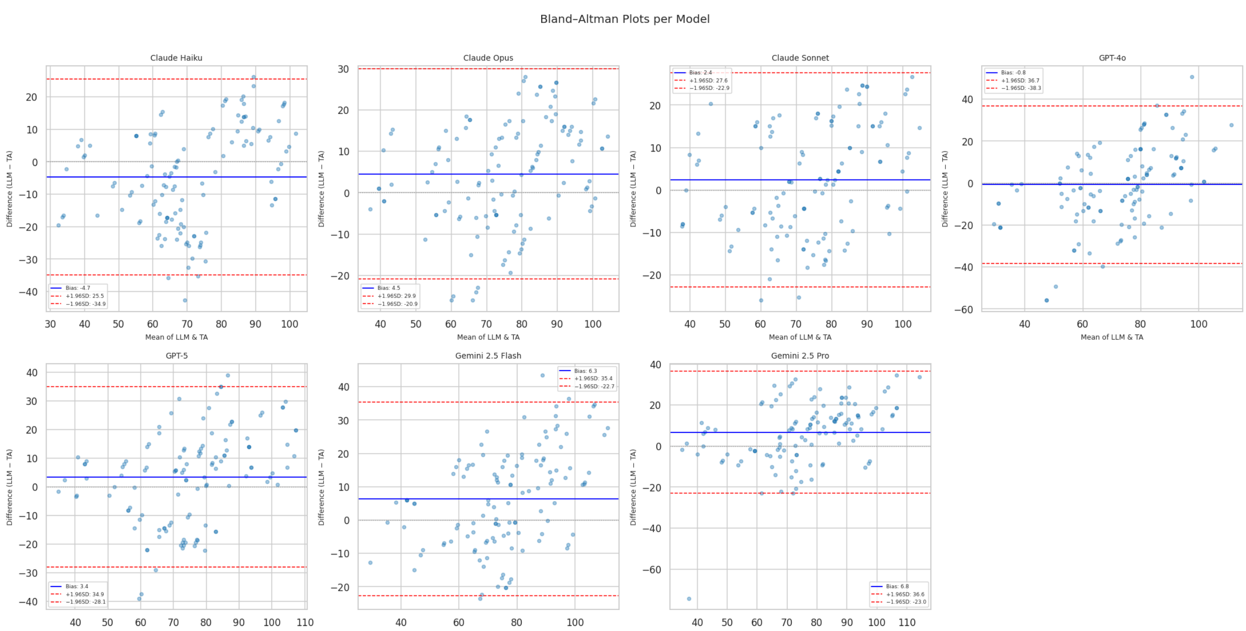

Figure 2. Bland–Altman agreement. The mean-difference lines sit below zero across the score range, confirming that the disagreement is a systematic offset and not a scale-dependent distortion.

5.4 The insertion test

Table 3 — ICC(2,1) with each model inserted as a fourth rater. Human-only baseline: ICC(2,1) = 0.956, Krippendorff's alpha = 0.956.

Model	ICC(2,1) with model	ICC(2,k)	Krippendorff's alpha	Change vs. human-only
Claude Opus	0.865	0.962	0.903	**−0.091**
Claude Sonnet	0.864	0.962	0.897	−0.092
Gemini 2.5 Pro	0.826	0.950	0.873	−0.130
Claude Haiku	0.824	0.949	0.847	−0.132
Gemini 2.5 Flash	0.815	0.946	0.872	−0.141
GPT-5	0.805	0.943	0.859	−0.151
GPT-4o	0.747	0.922	0.829	−0.209

No model preserves panel reliability. Adding the best available model to this teaching team costs 0.091 of ICC; adding the weakest costs 0.209. In the language a department would use: every one of these models is a *worse-than-average* member of this grading team, and replacing a TA with any of them measurably degrades the consistency of grading students receive.

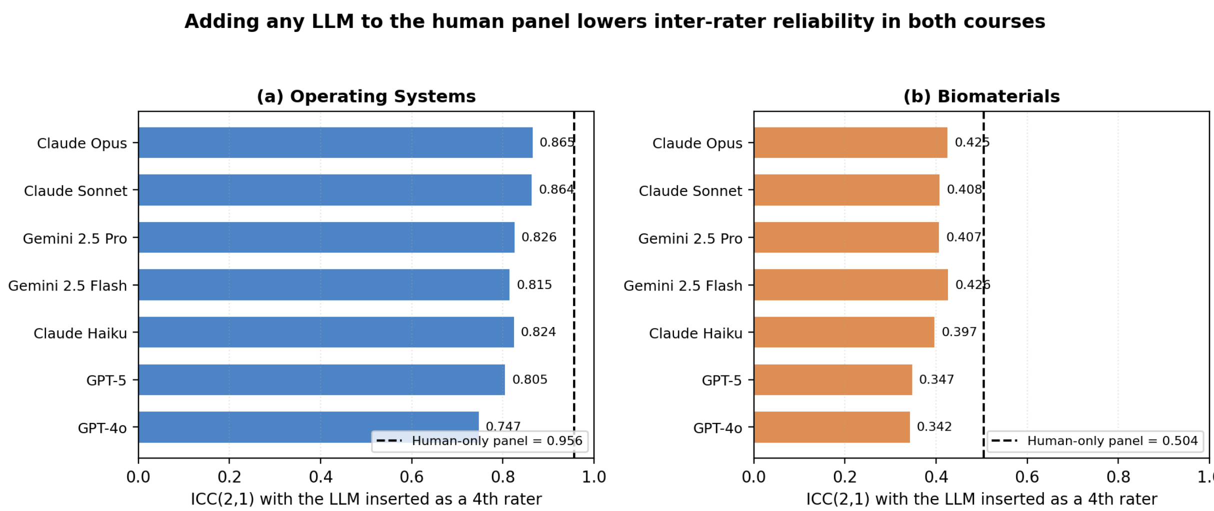

Figure 6. panel (a).

5.5 The within-human-variability test

Table 4 — Model disagreement with each individual TA, points per question. TA-TA ceiling = 1.113.

Model	vs. TA1	vs. TA2	vs. TA3	Mean	Ceiling ratio	Within human variability?
Claude Opus	3.485	2.787	3.074	**3.115**	**2.80x**	No
Claude Sonnet	3.707	2.887	3.169	3.254	2.92x	No
Gemini 2.5 Pro	3.874	3.103	3.461	3.480	3.13x	No
Gemini 2.5 Flash	3.947	3.347	3.640	3.644	3.27x	No
Claude Haiku	4.252	3.247	3.819	3.773	3.39x	No
GPT-5	4.149	3.581	3.883	3.871	3.48x	No
GPT-4o	5.060	4.645	4.770	4.825	4.34x	No

Zero of seven models fall within human variability. The gap is not marginal — the best model would need to reduce its disagreement by 64% to reach the ceiling.

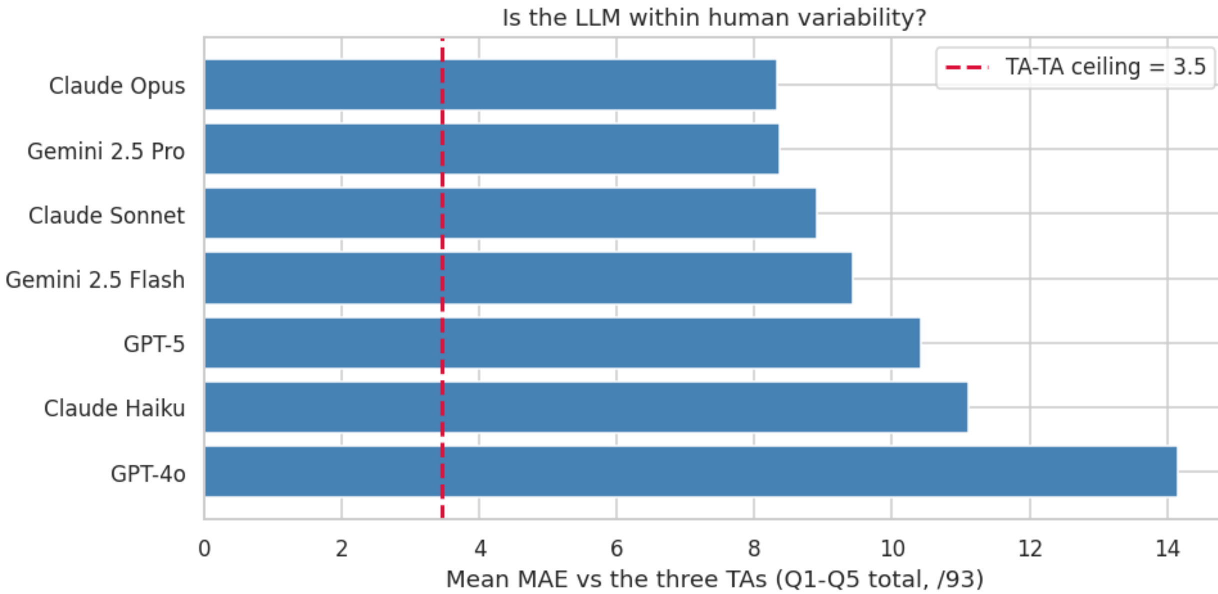

Figure 3.

5.6 Where the models fail, and how noisily

Per-question analysis locates the error. Q6 — the 40-point question, and the one TA2 did not grade — is the hardest for every model: Claude Haiku's MAE on Q6 is 7.76 points against 1.47 on Q4. Q4, the most constrained item, is where every model does best (MAE 1.31–1.58). The pattern is consistent with the pre-LLM autograding literature: constrained items automate well, open-ended synthesis does not.

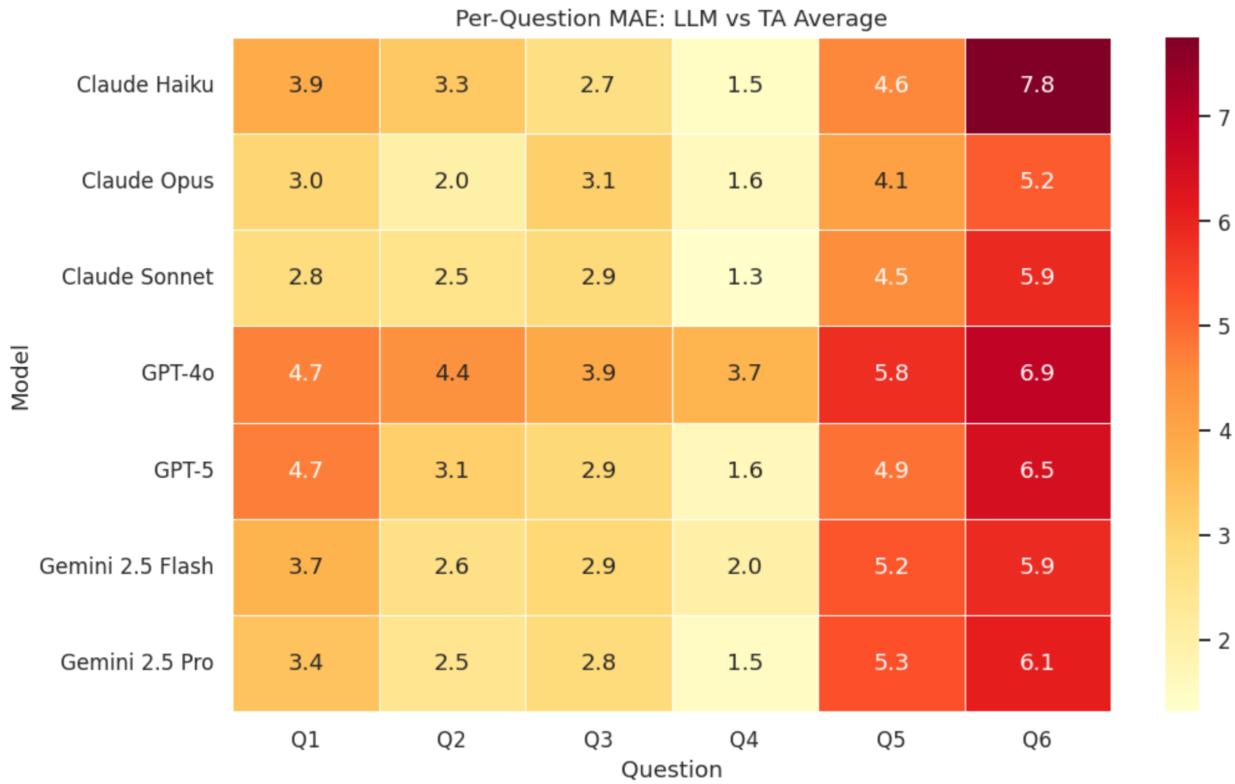

Figure 4.

Table 5 — Run-to-run standard deviation (points per question, mean over all student-question pairs).

Model	Run-to-run SD
Claude Opus	**0.765**
Claude Sonnet	0.883
Claude Haiku	0.994
GPT-5	1.027
GPT-4o	1.147
Gemini 2.5 Pro	1.376
Gemini 2.5 Flash	1.582

This is a fairness result, not just a reliability one. Gemini 2.5 Flash, invoked twice on identical work, produces scores differing by 1.58 points per question on average — 9.5 points across a six-question assignment. The models with the lowest noise are the two that also lead on accuracy, but the ordering is not identical: GPT-4o is mid-pack on noise despite being last on accuracy, and Gemini 2.5 Pro is third on accuracy but second-noisiest.

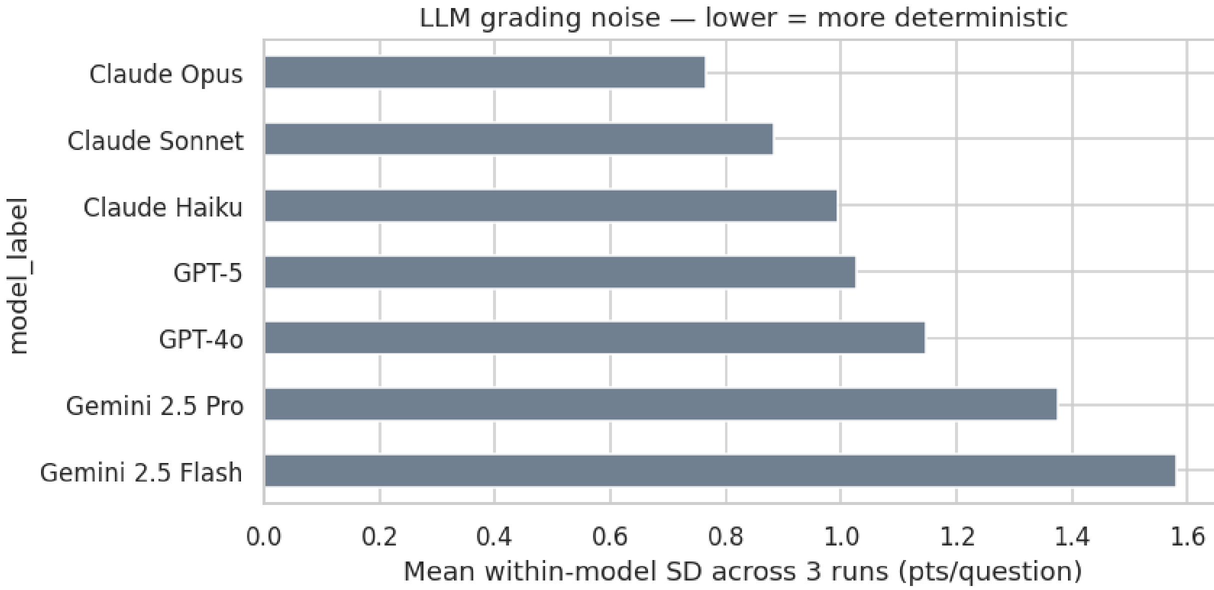

Figure 5.

5.7 What the conventional analysis would have concluded

We now run the analysis the conventional way on the same data: reference = the mean of the three recorded TA totals (inheriting the TA2/Q6 gap), and each (student, run) row treated as an independent observation (n = 120 rather than n = 40).

Table 6 — Naive vs. protocol-corrected bias, Operating Systems. Negative bias = harsher than the human consensus.

Model	Naive bias	Corrected bias	Sign flip?	Naive p	Corrected p (n=40)	Cohen's d	Correc
Claude Haiku	−4.70	−13.52	—	0.001	<0.0001	−0.908	Signific
GPT-4o	−0.75	−9.57	—	**0.667**	**0.0013**	−0.548	Signific
Claude Sonnet	**+2.35**	−6.46	**Yes**	0.047	0.0038	−0.530	Signific
GPT-5	**+3.39**	−5.42	**Yes**	0.023	0.0366	−0.359	Signific
Claude Opus	**+4.50**	−4.31	**Yes**	<0.001	0.0467	−0.366	Signific
Gemini 2.5 Flash	**+6.35**	−2.47	**Yes**	**<0.001**	**0.2162**	−0.195	Not sig
Gemini 2.5 Pro	**+6.76**	−2.06	**Yes**	**<0.001**	**0.3011**	−0.166	Not sig

The conventional analysis of this dataset would have supported the following claims, all of which are artefacts:

1. **That five of seven models grade *generously*.** They do not; all seven grade harshly. The sign reversal is driven entirely by the missing-TA2-Q6 reference contamination.
2. **That Gemini 2.5 Pro and Flash exhibit highly significant bias ($p < 0.001$).** Corrected, these are the two models whose bias is *not* statistically distinguishable from zero ($p = 0.30$ and $p = 0.22$) — they are the best-calibrated models in the study.
3. **That GPT-4o is unbiased ($p = 0.667$).** Corrected, GPT-4o is significantly harsh ($p = 0.0013$, $d = −0.548$).

Two distinct errors produce this. Reference contamination shifts every bias estimate by a constant. Pseudoreplication — treating three runs of the same student as three independent observations — inflates n threefold and shrinks p-values, manufacturing significance for effects that are small relative to between-student variance.

Neither error is exotic, and neither is visible from within the naive analysis. Only the step-1 requirement to characterise the human panel *before* scoring any model surfaces the first, and only run-averaging surfaces the second.

---

6. Study 2 — Biomaterials

6.1 A different ceiling

Table 7 — Inter-human agreement, Biomaterials, with Operating Systems for comparison.

Measure	Biomaterials	Operating Systems
ICC(2,1), single rater	**0.504** [0.220, 0.730]	0.956
ICC(2,k)	0.753	0.985
Krippendorff's alpha	0.469	0.956
Kendall's W	0.505	0.969
Mean pairwise weighted kappa	0.516	0.896
Mean human-human MAE	**0.227 pts/q** [0.133, 0.320]	1.113 pts/q
Human mean totals (/10)	H1 9.30, H2 7.80, H3 9.10	—

The same three-grader protocol, applied to a different course, yields a fundamentally different instrument. ICC(2,1) = 0.504 is moderate at best, and the graders' mean totals span 1.5 points out of 10 — H2 is a markedly harsher grader than H1 and H3.

Two factors contribute. The rubric is coarse: five questions at 2 points each, with most human scores falling in [1.5, 2.0]. This range restriction mechanically depresses ICC, which is a ratio of between-subject to total variance — when students genuinely differ little, even small rater disagreements dominate. It also drives the negative R^2 values in Table 8: with almost no variance around the consensus to explain, any deviation makes a model worse than predicting the mean. Under range restriction, R^2 is the wrong summary and MAE is the right one.

This is the point of including the study. Had we evaluated a model here against H2, it would look accurate; against H1, harsh. The ceiling framework makes the ambiguity explicit instead of hiding it in the choice of reference grader.

6.2 Models against the consensus

Table 8 — Question-level metrics vs. consensus (25 instances, 2 points max per question). Reference: human-human MAE = 0.227.

Model	MAE	MAE 95% CI	RMSE	Pearson r	Bias	Bias 95% CI	Cohen's d
Claude Haiku	**0.339**	[0.25, 0.44]	**0.416**	0.464	**−0.265**	[−0.39, −0.14]	−0.809

Model	MAE	MAE 95% CI	RMSE	Pearson r	Bias	Bias 95% CI	Cohen's d
Claude Opus	0.385	[0.28, 0.49]	0.465	0.671	−0.372	[−0.48, −0.26]	−1.310
Claude Sonnet	0.386	[0.29, 0.49]	0.461	0.635	−0.373	[−0.49, −0.27]	−1.348
Gemini 2.5 Flash	0.419	[0.29, 0.58]	0.554	**0.690**	−0.405	[−0.56, −0.27]	−1.054
Gemini 2.5 Pro	0.423	[0.27, 0.60]	0.594	0.634	−0.397	[−0.58, −0.23]	−0.880
GPT-4o	0.507	[0.41, 0.60]	0.558	0.616	−0.493	[−0.60, −0.39]	−1.857
GPT-5	0.544	[0.40, 0.70]	0.663	0.590	−0.524	[−0.68, −0.37]	−1.263

Every confidence interval on bias excludes zero. All seven models are significantly harsh (paired t and Wilcoxon both p 0.001), with large effect sizes (|d| = 0.81–1.86). On a 10-point assignment, the human graders average 8.7 and the models average 6.1–7.4 — a gap of well over a letter grade.

The MAE intervals overlap heavily. With 25 question instances, the ordering of adjacent models in this study is **not** statistically resolved, and we do not claim it is. What is resolved is the direction: every model, harsh, with a large effect size.

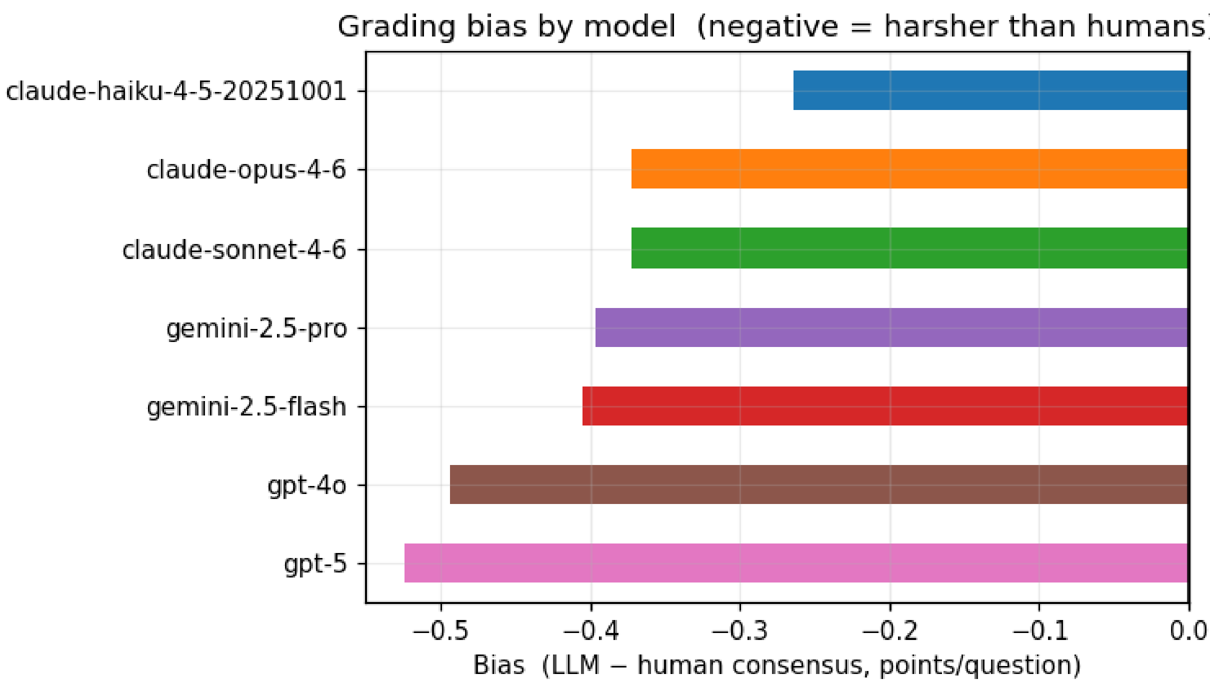

Figure 7.

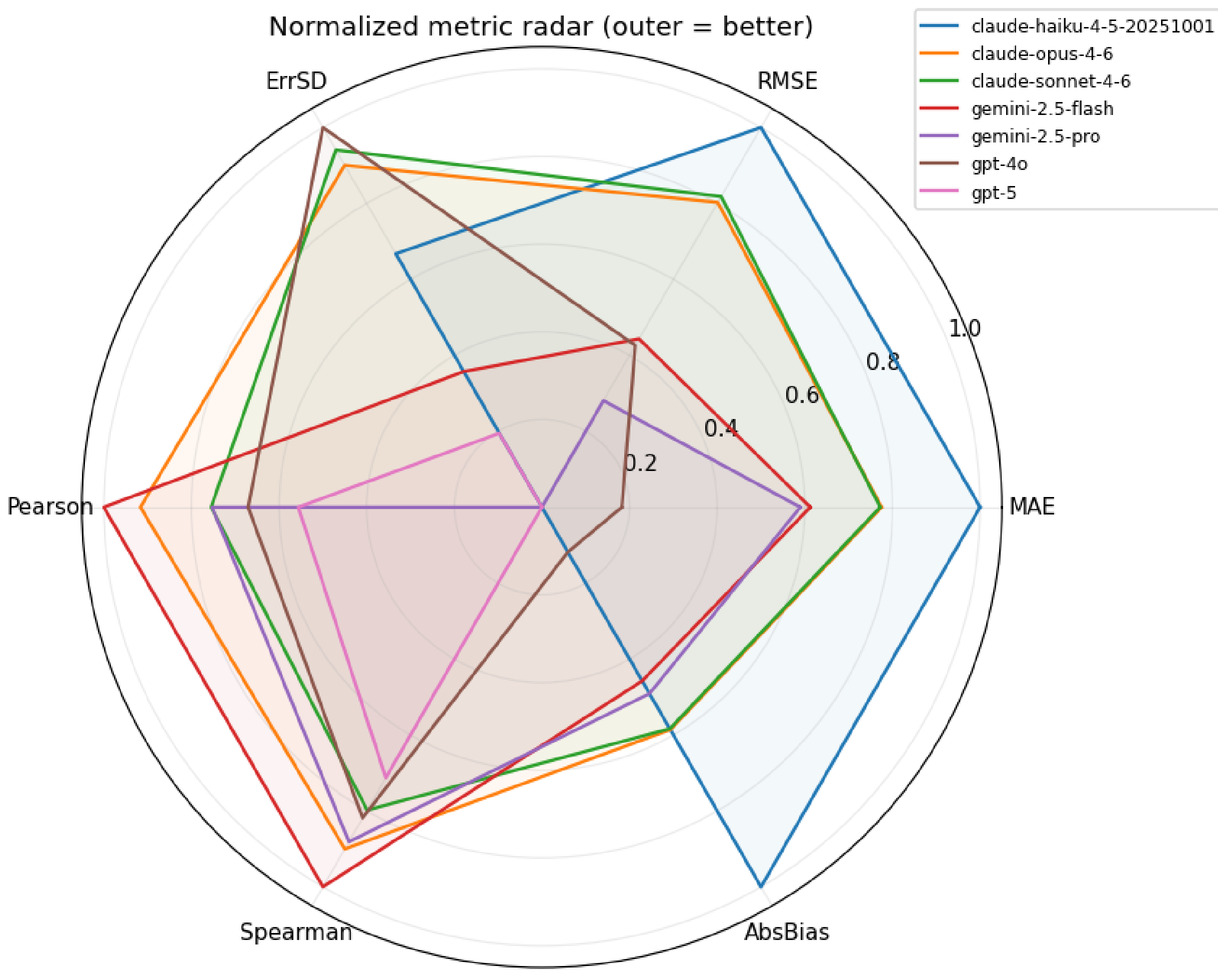

Figure 8.

6.3 Insertion, variability, and noise

The human-only baseline is ICC(2,1) = 0.504. With a model inserted as a fourth grader: Gemini 2.5 Flash 0.426, Claude Opus 0.425, Claude Sonnet 0.408, Gemini 2.5 Pro 0.407, Claude Haiku 0.397, GPT-5 0.347, GPT-4o 0.342 — drops of 0.077 to 0.161.

The verdict matches Operating Systems: no model preserves panel reliability, even against a panel that is itself only moderately reliable.

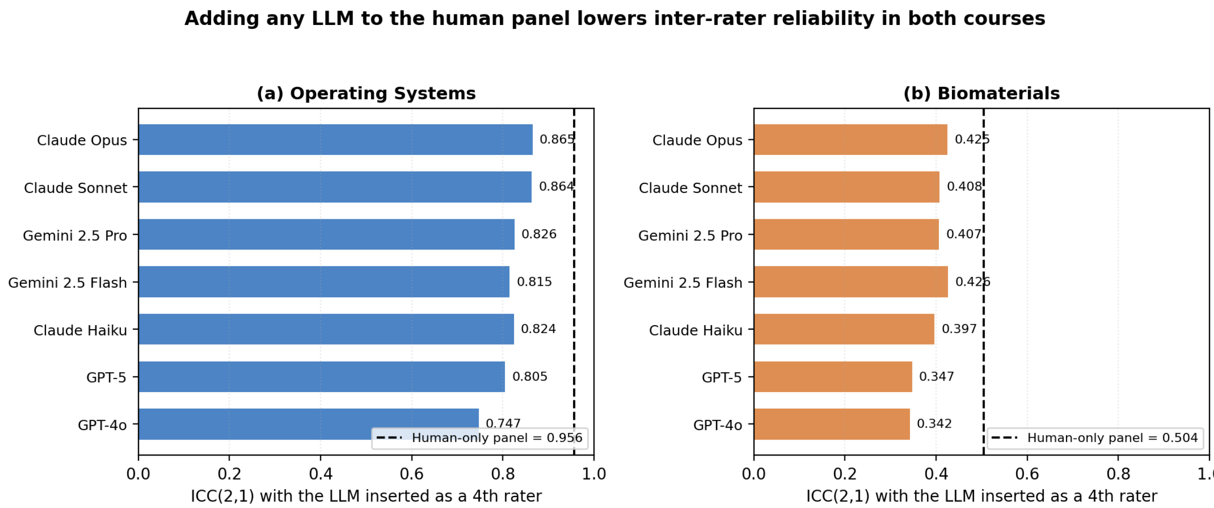

Figure 6. panel (b).

On the within-human test, model MAE against individual humans ranges from 0.399 (Claude Haiku) to 0.554 (GPT-5), against a ceiling of 0.227 — ceiling ratios of 1.76x to 2.44x. Again, zero of seven within human variability.

Run-to-run SD ranges from 0.066 (Claude Opus) to 0.208 (GPT-4o) points per question, on a 2-point scale — proportionally 3.3%–10.4% of the item's value. The ordering of models by noise is similar to Operating Systems: the Claude models are the most stable, GPT-4o and Gemini 2.5 Flash the least.

6.4 One instructive item

Per-question analysis flags Q4 as an *interesting failure*: the humans agreed on it more than on any other item (SD 0.115) and the consensus was 1.9/2.0, yet the models' mean absolute error was 0.444 — their second-worst. An item that is unambiguous to every human grader is one the models systematically misread. Conversely on Q5 the humans were most split (SD 0.273, consensus 1.5) while the models were confidently wrong in a consistent direction.

Both patterns argue against using model-human disagreement as an automatic flag for a contested item: the two do not track each other.

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7. Cross-Domain Synthesis

7.1 What replicates, stated precisely

The two courses differ in discipline, rubric granularity (2-point vs. 40-point items), scale (10 vs. 133 points), cohort size, input modality (PDF vs. structured text), and human ceiling (0.504 vs. 0.956). Against that, we ask what survives.

Rank correlation across the two courses (Spearman's rho, n = 7 models):

Metric	rho	p
Composite score	0.750	0.052
ICC as 4th rater	0.643	0.119
Run-to-run SD	0.679	0.094
MAE	0.571	0.180

The full ranking does not replicate. None of these correlations reaches significance at n = 7, and we explicitly decline the claim that our model ordering generalises. Reporting it as a stable leaderboard would not be supported by this data.

What does replicate:

1. **The top two, exactly.** Claude Opus is rank 1 and Claude Sonnet rank 2 in both courses, on every individual metric we computed.
2. **The OpenAI models are bottom-tier in both**, though they trade places — GPT-4o last on Operating Systems, GPT-5 last on Biomaterials.
3. **The middle does not order consistently.** Gemini 2.5 Pro is 3rd on Operating Systems and 5th on Biomaterials; Claude Haiku is 6th and 4th respectively.
4. **Both verdicts replicate without qualification.** Zero of seven models within human variability, in both courses. Every model harsh, in both courses — twelve of fourteen model-course pairs significantly so, the two exceptions being Gemini 2.5 Pro and Flash on Operating Systems.

7.2 The ceiling ratio

Table 9 — Ceiling ratios: model disagreement with humans, in units of human-human disagreement.

Model	Operating Systems	Biomaterials
Claude Opus	2.80x	1.80x
Claude Sonnet	2.92x	1.85x
Gemini 2.5 Pro	3.13x	1.98x
Gemini 2.5 Flash	3.27x	1.94x
Claude Haiku	3.39x	**1.76x**
GPT-5	3.48x	2.44x
GPT-4o	4.34x	2.31x

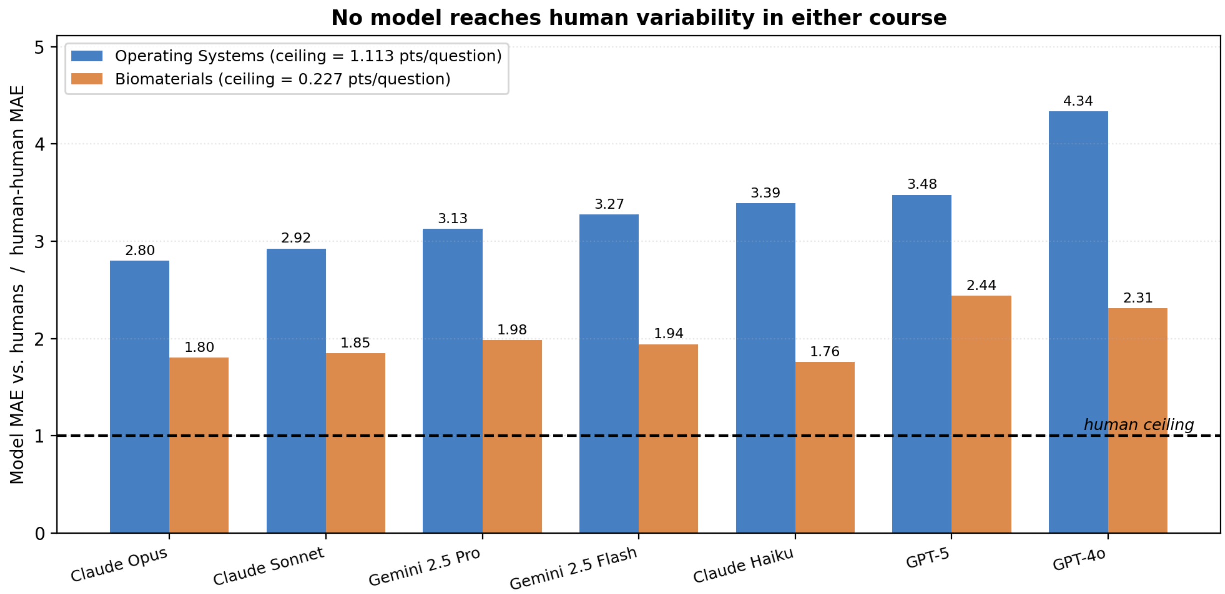

Figure 9.

Every bar in both courses is above 1.0. The ratios are *lower* in Biomaterials not because the models grade it better in absolute terms — they do not — but because the humans grade it less consistently, lowering the bar. This is the framework's central point rendered numerically: the same model can be 2.80x or 1.80x from acceptable depending entirely on whose grading it is being asked to match.

It also implies a practical corollary. **The courses where an LLM grader is most likely to clear the human bar are exactly the courses where human grading is least reliable** — which is a much weaker endorsement than "the LLM is accurate", and should be described honestly as such when deploying.

7.3 Bias dominates variance

Across both courses, the signed bias accounts for most of the mean absolute error. On Biomaterials, mean bias is −0.40 points/question against a mean MAE of 0.43 — the models are almost *entirely* offset, with comparatively little residual scatter. On Operating Systems the offset is a smaller share of the total, but is still statistically significant for five of seven models.

This is the most actionable finding in the paper, and Section 8.1 develops it.

7.4 Latency

Table 10 — Grading latency per submission (seconds).

Model	OS (median)	Biomaterials (mean)
GPT-4o	**8.2**	**7.4**
Gemini 2.5 Flash	25.9	28.5
Claude Haiku	27.0	20.0
Gemini 2.5 Pro	37.6	31.3
Claude Opus	52.6	37.9

Model	OS (median)	Biomaterials (mean)
Claude Sonnet	54.2	38.8
GPT-5	134.2	88.0

Latency spans a factor of 16 and is *inversely* related to accuracy at both extremes: the fastest model is the least accurate, and the slowest (GPT-5) is mid-pack. Claude Opus delivers the best accuracy at roughly 40% of GPT-5's latency. For a 40-student assignment, even the slowest configuration completes a full grading pass in under 90 minutes unattended, so latency is unlikely to be the binding constraint in practice.

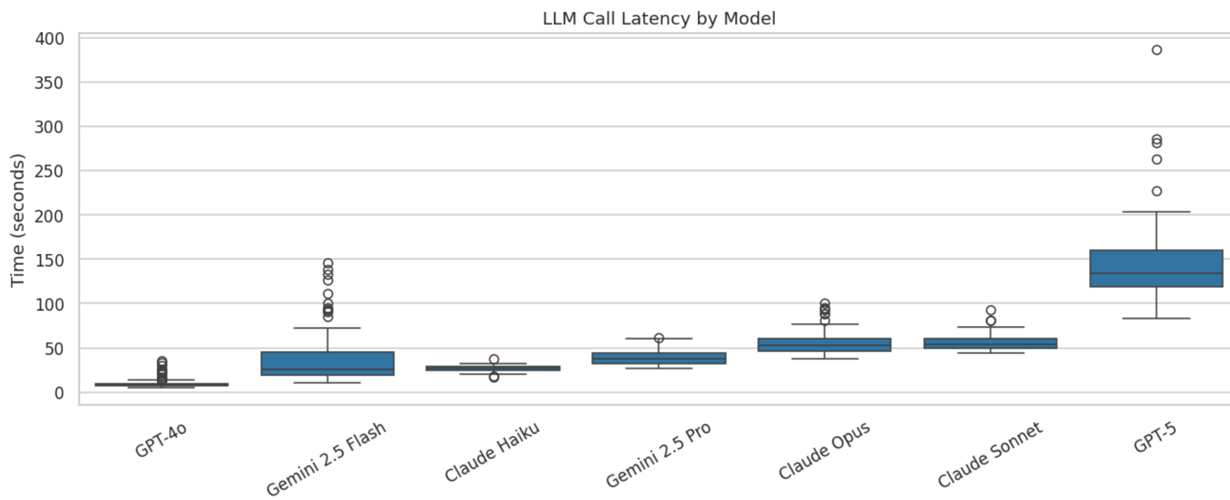

Figure 10.

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8. Discussion

8.1 The error is an offset, and offsets are correctable

The strongest practical implication follows from Section 7.3. A grader that is wrong at random is unfixable without improving the grader. A grader that is *consistently* wrong in one direction is a calibration problem.

On Biomaterials, subtracting each model's mean bias from its scores would reduce mean absolute error from roughly 0.43 to roughly the residual error SD — the models' rank ordering of students is already reasonable, and the harshness offset is what puts the absolute grades outside acceptable range. A per-rubric calibration constant, estimated from a modest sample of double-graded submissions, addresses this directly.

We emphasise the constraints. The offset is *per rubric*, not per model: Claude Haiku's bias is −2.253 points/question on Operating Systems and −0.265 on Biomaterials, and these are not convertible into each other by rescaling. Calibration therefore requires human-graded submissions from the same assignment, which reintroduces exactly the human effort automation was meant to remove — though at a sample size far below full grading. And calibration cannot repair the run-to-run noise of Section 5.6, which is a separate defect.

8.2 What we do not recommend

Autonomous grading of record. Not supported by this data, in either course. Zero of fourteen model-course pairs fall within human variability, and every model degrades panel reliability when inserted. The gap is large, not marginal.

Substituting a model for a TA on a grading team. This is precisely what the insertion test evaluates, and every model fails it.

Trusting a high correlation. Table 2 is the cautionary case: r = 0.879 alongside an MAE 2.80x the human ceiling. Correlation measures whether the model ranks students correctly; it is nearly blind to a uniform offset. A model can correlate at 0.88 with the TAs while awarding every student a grade one band too low. Any evaluation reporting correlation without absolute error, referenced to human-human disagreement, is under-reporting.

8.3 What the data does support

Second-opinion flagging. Use the model as an additional, non-authoritative rater and surface submissions where model and human diverge most. The model does not need to be correct for this to be valuable — it needs to be uncorrelated enough with the human's error to catch slips. Section 6.4 is a caution here: model-human divergence did not track item contentiousness, so this should be validated per course rather than assumed.

Triage for grading order. Model scores correlate 0.77–0.88 with the consensus, which is ample for ordering a grading queue so that borderline submissions reach a human first.

Draft feedback. The models produce per-question strengths, weaknesses, and breakdowns. Nothing in our data speaks to the quality of that prose — we evaluated only scores — but the grading task and the feedback task fail differently, and a model too harsh by a consistent offset may still write useful formative comments. **This needs separate evaluation.**

Rubric quality diagnosis. The Biomaterials ceiling of 0.504 was discovered by running step 1 of the protocol. That finding is independently valuable to the instructor, irrespective of any LLM: it says the rubric does not discriminate reliably between graders. Running the human half of this protocol is worthwhile even if no model is ever deployed.

8.4 Model selection, if deploying

Claude Opus 4.6 and Claude Sonnet 4.6 lead on every metric in both courses and are the two most run-stable models. Sonnet achieves near-identical accuracy to Opus at comparable latency. GPT-4o was last or near-last on accuracy in both courses, and its speed advantage does not compensate. Gemini 2.5 Pro and Flash were the *best-calibrated* models on Operating Systems (the only two with statistically non-significant bias) but the noisiest across runs — a combination that suits ensemble averaging, where repeated sampling suppresses noise while good calibration is retained. We did not test ensembling, and flag it as the most promising untested configuration suggested by our results.

8.5 Reconciling the contradictory literature

Section 2.2 set out an unresolved disagreement: Gobrecht et al. report an automated grader beating human re-graders by 44% on median absolute error [3], Tang et al. report parity with human inter-rater reliability [9], while Mathew et al. report QWK below 0.30 against a human ceiling of 0.72 [10], and Caraeni et al. judge accuracy too low for deployment [11].

Our two studies reproduce that disagreement *within a single paper, using one protocol and one set of models.* Ranked against the Operating Systems TAs, the best model sits 2.80x outside human variability and looks clearly unfit. Ranked against the Biomaterials graders, the same model sits 1.80x outside — still failing, but by a margin that a slightly different rubric or a slightly noisier panel would erase. The models did not change. The graders did.

This suggests a specific reading of the literature. Studies reporting parity or better tend to involve reference conditions with high human variability: re-grading historic exams without the original grading context [3], or rubrics on which instructors themselves diverge. Studies reporting failure tend to involve reference conditions with well-controlled human agreement — essay corpora with trained, calibrated raters reaching QWK 0.72 [10]. **The apparent disagreement about model capability may be substantially a disagreement about reference quality.**

We advance this as an explanation consistent with our data, not as a demonstrated one: we cannot recompute other authors' ceilings, and we do not claim their conclusions are wrong. The testable prediction is that reported LLM-human agreement should correlate negatively with the inter-human reliability of each study's reference panel. That prediction could be checked by a meta-analysis, and we suggest it as future work.

The practical implication is uncomfortable but important, and it restates Section 7.2: an LLM grader is most likely to clear the human bar precisely where human grading is least reliable. ”The model performs as well as our graders” and ”our graders do not agree with each other” are compatible statements, and only the first tends to get reported.

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9. Threats to Validity

The prompt is not identical across providers. The grading service emits a compact delimiter-based format for Gemini models, including an explicit instruction to keep each field under 30 words, and a JSON format for OpenAI and Anthropic models. The Gemini results are therefore confounded with a different prompt and a tighter output-length constraint. Gemini's cross-model comparisons should be read with this in mind; the within-Gemini findings (calibration, noise) are unaffected.

Decoding settings are not uniform. Six models ran at temperature 0.1; GPT-5 ran with high reasoning effort and no temperature control. GPT-5's run-to-run SD is therefore not measured under the same conditions as the others, and its noise figure is not directly comparable.

Token budgets differ — 2000 max tokens for Anthropic and Gemini models against 16384 for non-reasoning OpenAI models — which could truncate long rubric feedback asymmetrically.

Input modality is confounded with course. Operating Systems answers were supplied as structured text; Biomaterials submissions as directly-ingested PDFs. Any cross-course difference may be a modality effect rather than a course effect. This does not affect the within-course results, which are the basis for all our primary claims.

The Biomaterials study is underpowered. Five students, 25 question instances, wide overlapping confidence intervals. We treat it as a contrast case for the protocol, not as independent replication, and we do not claim its model ordering is resolved.

The consensus is a reference, not ground truth. Averaging three graders does not produce correctness. Where the humans agree at ICC 0.504, the consensus is itself an unreliable target, and a model penalised against it may not be wrong.

Range restriction on Biomaterials. Five 2-point items with most scores in [1.5, 2.0] mechanically depresses ICC and produces negative R^2. We report MAE alongside for this reason, but the ICC comparison across the two courses should not be read as a pure difference in grader skill.

One assignment per course, one institution. Both studies are single assignments from a single institution with a single rubric each. Question-level findings (Section 5.6) in particular may not transfer.

Human graders were not blinded to the study, and grading order was not randomised.

Model versions are a snapshot. These are specific model versions accessed over a bounded period; provider-side updates can change behaviour without notice. Absolute numbers should be treated as dated; the protocol is what we intend to be durable.

Cost was not measured. We report latency but not API cost per submission, which is likely a binding constraint at scale and which varies by more than an order of magnitude across the models tested.

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10. Conclusion

We evaluated seven frontier language models as rubric graders on two engineering assignments, each graded independently by three humans, with every model run three times — 945 model gradings in total.

Our central methodological claim is that **inter-human agreement must be measured before any claim about LLM grading accuracy is interpretable**, and we demonstrated this rather than asserting it. On the same Operating Systems data, a conventional single-reference, run-pooled analysis reverses the sign of measured bias for five of seven models, reports highly significant bias for the two best-calibrated models, and reports no significant bias for a model that is in fact significantly harsh. The two courses' human ceilings differ by a factor of nearly two in ICC, so there is no universal bar an LLM grader can be held to. This also gives us a candidate explanation for why the published literature reaches opposite verdicts on LLM grading: with the models and protocol held fixed, changing only the human panel moves the best model from 2.80x to 1.80x outside human variability. We suggest the field's disagreement is substantially about reference quality, and note the meta-analytic prediction that would test it.

Our central empirical claim is that **no model tested reaches human variability in either course.** The best model disagrees with the human panel 2.80x as much as the humans disagree with each other on Operating Systems and 1.76x on Biomaterials, and inserting any model as an additional rater lowers panel reliability in both. This holds despite Pearson correlations of up to 0.88, which we take as evidence that correlation alone is an inadequate reporting standard for grading applications.

Our central practical claim is that **the dominant failure is a systematic harshness offset rather than random error** — twelve of fourteen model-course pairs are significantly harsh — and that this is correctable by per-rubric calibration in a way that random error would not be.

This supports assistive deployments (triage, second-opinion flagging, calibrated first-pass scoring under human review) and does not support autonomous grading of record.

We recommend that studies of LLM grading report, as a minimum: inter-human reliability for the same items, absolute error expressed relative to human-human disagreement, and run-to-run variability across repeated invocations. The first is the ceiling, the second is the honest accuracy measure, and the third is a fairness property that single-run studies cannot see.

Reproducibility

All analyses derive from two Jupyter notebooks applying the identical protocol: `Grading_Dataset_OS/llm_vs_huma` (Phase 9 implements the human-ceiling protocol) and `Biomaterials/multiLLM_multiHuman_analysis.ipynb`. The grading pipeline is `grading_service.py`; the experiment drivers are `Grading_Dataset_OS/test_grading_rer` and `Biomaterials/grade_biomaterials.py`. Figures 6 and 9 are produced by `docs/paper/make_paper_figures` Grades, both human and model, are in `Grading_Dataset_OS/consolidated_results_rerun_4/` and `Biomaterials/grading_results/`.

Acknowledgements

[ACKNOWLEDGEMENTS]

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